MATH-334 Numerical Analysis

Error & Root Finding Problem

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Recommended Books

Text Book:

(a) Numerical Analysis (8^{th} ed) By Burden & Faires

Reference Books:

- Numerical Methods for Engineers (6^{th} Edition)Steven C. Chapra Raymond P. Canale. McGraw-Hill Companies, Inc., 1221 Avenue of the Americas, New York, NY 10020.
- E. Kreyszing. Advanced Engineering Mathematics 9^{th} ed.

Outline

Error Summary

True Error	E_t=true value $-$ approximated value
Absolute Error	$\lvert E_t \rvert$
Percentage Relative Error	$\epsilon_t = \left(\dfrac{\text{true value} - \text{appr. value}}{\text{true value}} \right) \times 100$

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Stopping Criterion	$\lvert \epsilon_a \rvert \leq \epsilon_s$
Tolerance	$\epsilon_s = \left(0.5 \times 10^{2-k}\right) \%$

Error criterion for k significant figure

$$\epsilon_s = \left(0.5 \times 10^{2-k}\right) \%, \qquad \text{(Scarborough, 1966).}$$

Root Finding Problems

Many problems in Science and Engineering are expressed as:

$$\boxed{\text{Given a continuous function } f(x), \text{ find the value } p \text{ such that } f(p) = 0}$$

These problems are called root finding problems.

Solution Methods

Several ways to solve nonlinear equations are possible:

- Analytical Solutions
  - Possible for special equations only
- Graphical Solutions
  - Useful for providing initial guesses for other methods
- Numerical Solutions
  - Open methods
  - Bracketing methods

Numerical Methods

Many methods are available to solve nonlinear equations:

- Bisection Method→ bracketing method
- Newton's Method→ open method
- Secant Method →open method
- False position Method (bracketing method)
- Fixed point iterations (open method)
- Muller's Method
- Bairstow's Method
- ....

Bracketing Methods

- In bracketing methods, the method starts with an interval that contains the root and a procedure is used to obtain a smaller interval containing the root.

- Examples of bracketing methods:
  - Bisection method
  - False position method (Regula-Falsi)

Open Methods

- In the open methods, the method starts with one or more initial guess points. In each iteration, a new guess of the root is obtained.
- Open methods are usually more efficient than bracketing methods.
- They may not converge to a root.

Outline

Bisection Algorithm

Assumptions:

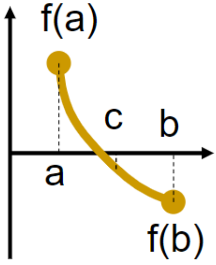
- f(x) is continuous on [a,b]
- $f(a)\, f(b) < 0$

Algorithm:

Loop

1. Compute the mid point $c=(a+b)/2$
2. Evaluate f(c)
3. If $f(a)\, f(c) < 0$ then new interval [a, c]
   If $f(a)\, f(c) > 0$ then new interval [c, b]

End loop



Outline

Newton-Raphson's Method

The Eq. of tangent line passing through $(x_0, f(x_0))$ is

$$f(x) - f(x_0) = \underbrace{f'(x_0)}_{m}(x - x_0)$$

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$$-f(x_0) = f'(x_0)(x_1 - x_0)$$

Simplification gives

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

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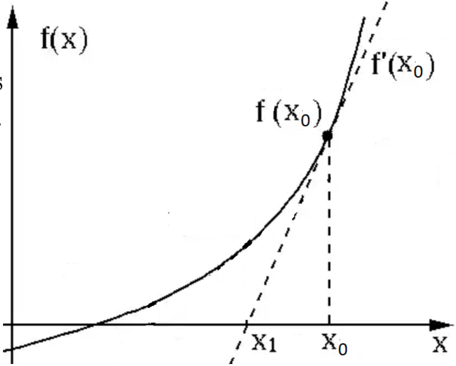

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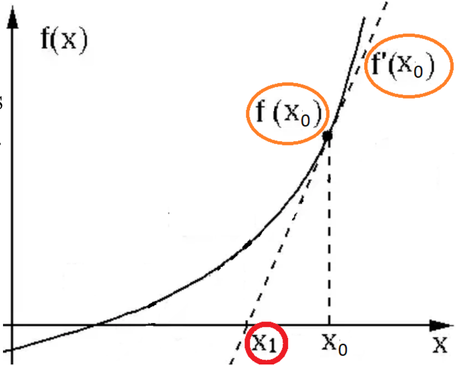
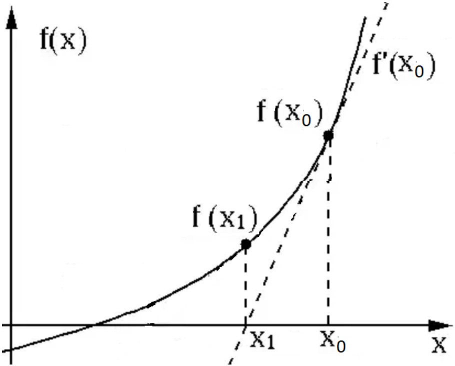
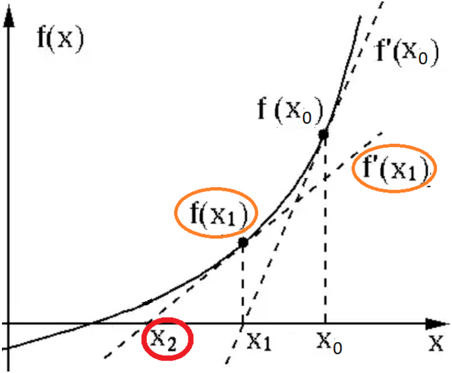


Newton-Raphson's Method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}, \quad \text{provided} \quad f'(x_1) \neq 0.$$





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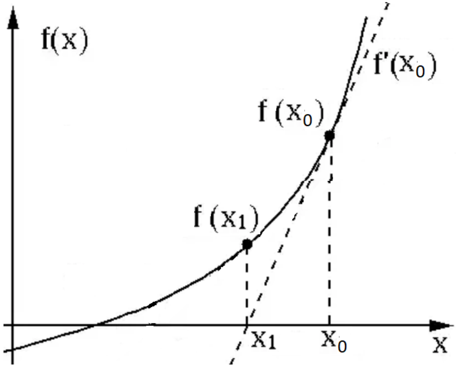
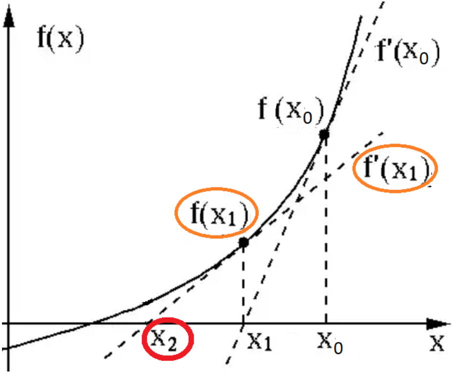
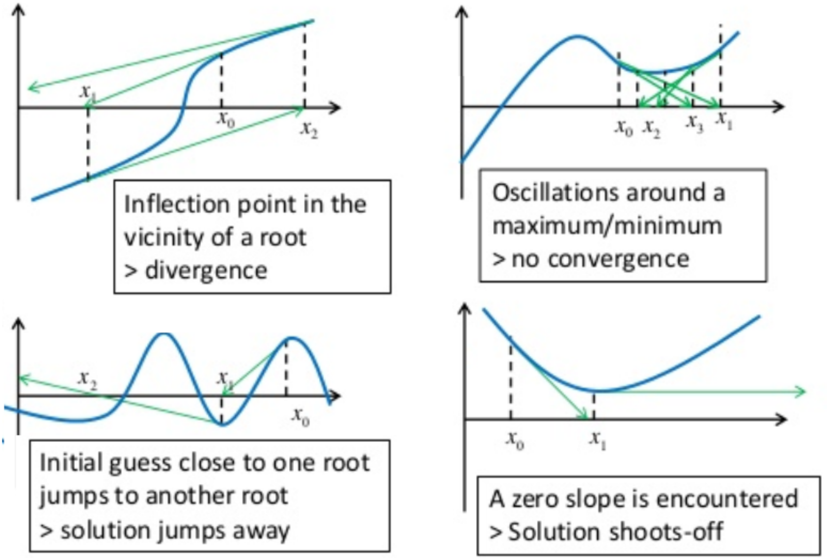




$$x_k = x_{k-1} - \frac{f(x_{k-1})}{f'(x_{k-1})}, \quad \text{provided} \quad f'(x_{k-1}) \neq 0.$$

Pitfall of NR method



Determine the positive root of $f(x) = x^{10} - 1$ using the Newton-Raphson method and an initial guess of $x = 0.5$.

Outline

Secant Method

The equation of Secant line for the function $f(x)$ passing through $(x_0, f(x_0))$ and $(x_1, f(x_1))$ is given by

$$\frac{f(x)-f(x_1)}{x-x_1}=\frac{f(x_1)-f(x_0)}{x_1-x_0}$$

The secant line intersect the $x-axis$ at $x=x_2$ assuming $f(x_2)\simeq 0$ gives

$$\begin{aligned}\frac{0-f(x_1)}{x_2-x_1}&=\frac{f(x_1)-f(x_0)}{x_1-x_0}\\ \frac{x_2-x_1}{-f(x_1)}&=\frac{x_1-x_0}{f(x_1)-f(x_0)}\end{aligned}$$

$$x_2=x_0-\frac{(x_1-x_0)\,f(x_0)}{f(x_1)-f(x_0)} \quad \text{and in general,}$$

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$$\frac{0 - f(x_1)}{x_2 - x_1} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$\frac{x_2 - x_1}{-f(x_1)} = \frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

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$$x_k = x_{k-1} - \frac{(x_{k-2} - x_{k-1})\, f(x_{k-1})}{f(x_{k-2}) - f(x_{k-1})} \quad \text{or} \quad x_k = \frac{x_{k-1} f(x_{k-2}) - x_{k-2} f(x_{k-1})}{f(x_{k-2}) - f(x_{k-1})}$$

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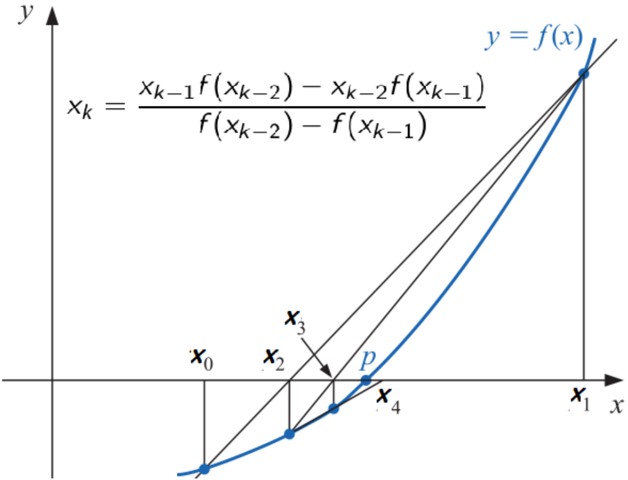

Outline

The method of False Position (also called Regula Falsi)

generates approximations in the same manner as the Secant method, but it includes a test to ensure that the root is always bracketed between successive iterations.

First choose initial approximations x_0 and x_1 with

$$f(x_0)f(x_1) < 0.$$

The approximation x_2 is chosen in the same manner as in the Secant method.

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(a) If $f(x_2)f(x_1) < 0$, then x_1 and x_2 bracket a root. Choose x_3 as the x-intercept of the line joining $(x_1, f(x_1))$ and $(x_2, f(x_2))$.

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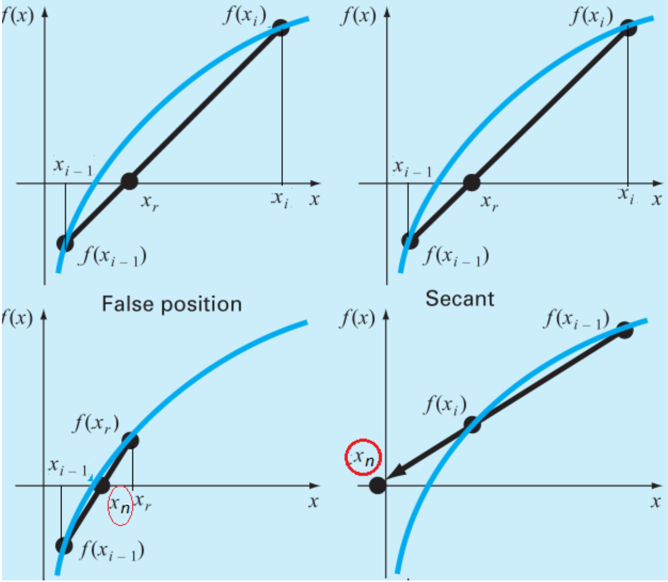
(b) If not, choose x_3 as the x-intercept of the line joining $(x_0, f(x_0))$ and $(x_2, f(x_2))$, and then interchange the indices on x_0 and x_1.


f(x)
f(x_i)
x_{i-1}
x_r
x_i
x
f(x_{i-1})
False position
f(x)
f(x_i)
x_{i-1}
x_r
x_i
x
f(x_{i-1})
Secant
f(x)
f(x_r)
x_{i-1}
x_n
x_r
x
f(x_{i-1})
f(x)
f(x_{i-1})
f(x_i)
x_n
x

Find a root for $\cos x - x + 2 = 0$ when $x_0 = 1$ and $x_1 = 2$

$$x_k = \frac{x_{k-1} f(x_{k-2}) - x_{k-2} f(x_{k-1})}{f(x_{k-2}), -f(x_{k-1})}$$

$f(x_0) = 1.5403, f(x_1) = -0.41615$

Secant Method:

$$x_2 = \frac{x_1 f(x_0) - x_0 f(x_1)}{f(x_0) - f(x_1)}$$

$x_2 = 1.78729$

$$x_3 = \frac{x_2 f(x_1) - x_1 f(x_2)}{f(x_1) - f(x_2)}$$

$x_3 = 1.78621$

$$x_4 = \frac{x_3 f(x_2) - x_2 f(x_3)}{f(x_2) - f(x_3)}$$

$x_4 = 1.78623$

Method of False Position:

$$x_2 = \frac{x_1 f(x_0) - x_0 f(x_1)}{f(x_0) - f(x_1)}$$

$x_2 = 1.78729, f(x_2).f_(x_0) < 0$

$$x_3 = \frac{x_2 f(x_0) - x_0 f(x_2)}{f(x_0) - f(x_2)}$$

$x_3 = 1.78622, f(x_3).f(x_2) < 0$

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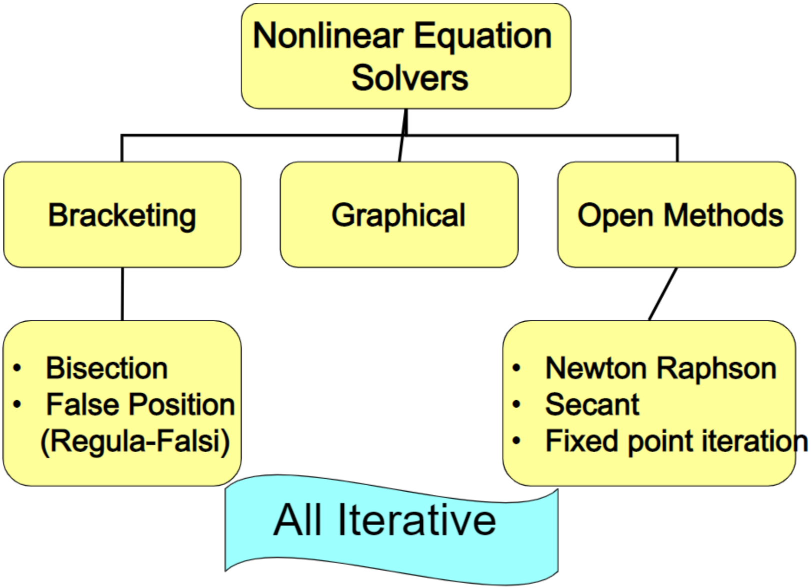
$x_4 = 1.78623$

Find a root of the equation

$$x = \cos x, \qquad [0.5, \frac{\pi}{4}]$$

	False Position	Secant	Newton
n	p_n	p_n	p_n
0	0.5	0.5	0.7853981635
1	0.7853981635	0.7853981635	0.7395361337
2	0.7363841388	0.7363841388	0.7390851781
3	0.7390581392	0.7390581392	0.7390851332
4	0.7390848638	0.7390851493	0.7390851332
5	0.7390851305	0.7390851332	
6	0.7390851332		

1. $4x = \sin x - e^x$
2. $\cos x - xe^x$
3. $x^3 + 4x^2 - 10$


Nonlinear Equation Solvers
Bracketing
Graphical
Open Methods
• Bisection
• False Position (Regula-Falsi)
• Newton Raphson
• Secant
• Fixed point iteration
All Iterative

Applications of Root Finding

Electrical Engineering:

Find **resistance,*R***, of a circuit such that the charge reaches q at specified time t

$$f(R) = e^{-Rt/2L} \cos\left[\left(\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}\right) t\right] - \frac{q}{q_0} = 0$$

L = inductance, C = capacitance,
q_0 = initial charge

Applications of Root Finding

Civil Engineering:

Find **horizontal component of tension, *H*,** in a cable that passes through y_0 and (*x*, *y*)

$$f(H) = \frac{H}{w}\left[\cosh\left(\frac{wx}{H}\right) - 1\right] + y_0 - y = 0$$

w = weight per unit length of cable

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